

LESSON 6-6 • SESSION 1 • PRACTICE • FORM A

Simplifying to Show Equivalence

Learning target – I can simplify an expression by combining like terms to test whether two expressions are equivalent.
6.EE.A.3

REMEMBER

- **Like terms** have the same variable part: $4x$ and $9x$.
 $4x$ and 4 are **not** like terms.
- Combine like terms by adding or subtracting **coefficients**: $4x + 9x = 13x$. A lone x is $1x$.
- Expressions are **equivalent** if they simplify to the same expression.

WORKED EXAMPLE

Simplify $7c + 9 + 6c$. Is it equivalent to $13c + 9$?

Step 1 Like terms: $7c$ and $6c$.

Step 2 Combine: $7c + 6c = 13c$. The 9 stays alone.

Step 3 $7c + 9 + 6c = 13c + 9$.

Answer $13c + 9$; yes, equivalent

WORDS like terms (*términos semejantes*) • combine (*combinar*) • equivalent expressions (*expresiones equivalentes*)

PRACTICE

show your work • label units

1. Simplify each expression.

a. $5x + 8 + 3x =$ _____

b. $9m + 4 - 2m =$ _____

c. $4x + 12 + 3x + 5 =$ _____

d. $9r + 7 + 6r + 8 =$ _____

e. $5n + 5n =$ _____

f. $6 + 2k + k =$ _____

2. **Supply Order.** Two students describe the same supply order. Ana writes $3p + 10 + 2p$. Luis writes $5p + 10$.

a. Name the like terms in Ana's expression. _____

b. Simplify Ana's expression. _____

c. Does it match Luis's expression? _____

d. Why can $3p$ and 10 not be combined? _____

3. **Simplify Fully.** Simplify $10m + 4 + 2m + 14$. 4. **Simplify Fully.** Simplify $8h + 21 + 3h - 6$.

Hint Group the m terms, then the constants.

Hint $21 - 6$ for the constants.

Show your work.

Show your work.

Simplified _____

Simplified _____

5. **Rashid's Claim.** Rashid says $4t + 16$ and $20t$ are equivalent because $4 + 16 = 20$. Justify or correct his claim.

Hint Do $4t$ and 16 have the same variable part?

Show your work.

Equivalent? _____

- 6. Three Expressions.** A is $2a + 3a + 10$. B is $5a + 10$. C is $5 + a + 10$. Simplify all three. Which two are equivalent?

Hint Simplify each one, then line them up.

Show your work.

Equivalent pair _____

- 7. Two Store Bills.** Bill A is $5p + 3 + 2p$. Bill B is $7p + 3$, where p is the price of one pen.

- a. Simplify Bill A. _____
- b. Are the two bills equivalent? _____
- c. Evaluate both bills when $p = 4$. What does this show? _____

- 8. Not Equivalent.** Are $4y + 5$ and $9y$ equivalent? Explain.

Hint Can $4y$ and 5 be combined?

Show your work.

Equivalent? _____

FIND THE ERROR

explain the mistake, then fix it

- 9. Willa's Work.** Willa simplified $12x + 3 + 4x$ and wrote $19x$, saying she combined all three terms. Explain her mistake. Then simplify correctly.

TEST PRACTICE

choose the best answer

- 10.** Simplify $6n + 4 + 2n$.

A $12n$ **B** $8n + 4$ **C** $6n + 6$ **D** $10n$

- 11.** Which pair are NOT like terms?

A $3x$ and $7x$ **B** $2y$ and y **C** $5m$ and 5 **D** $4k$ and $10k$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

30m Q	31 S	7p + 3 M	26h Z	20t X	11h + 15 R	5p + 10 T	12m + 18 E
		#2b	#3	#4	#7a	#7c	

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____